

RESEARCH ARTICLE

Diagnosing osteo-articular tuberculosis and multidrug resistance using real-time polymerase chain reaction and high-resolution melt-curve analysis

Kusum Sharma¹ | Megha Sharma²  | Aman Sharma³ | Mandeep Singh Dhillon⁴¹Department of Medical Microbiology, Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh, India²Department of Microbiology, All India Institute of Medical Sciences (AIIMS), Bilaspur, Himachal Pradesh, India³Department of Internal Medicine, Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh, India⁴Department of Orthopaedics, Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh, India**Correspondence**Kusum Sharma, Department of Medical Microbiology, Post Graduate Institute of Medical Education and Research (PGIMER), Chandigarh 160012, India.
Email: sharmakusum9@yahoo.co.in**Abstract**

The study evaluated real-time quantitative polymerase chain reaction (qPCR) and high-resolution melt-curve analysis (HRM) for simultaneous diagnosis of osteo-articular tuberculosis (OATB) and drug resistance. Two hundred and fifty synovial fluid and pus specimens (20 confirmed OATB by culture, 130 suspected OATB, and 100 controls) processed in the Department of Medical Microbiology, PGIMER were subjected to qPCR using *rpoB*, *MPB64*, and *IS6110* genes. All OATB positive specimens were subjected to HRM for detecting resistance to rifampicin and isoniazid. qPCR detected 129/150 OATB cases with a sensitivity of 86% (95% for confirmed and 84.6% for suspected OATB cases) and specificity of 100%. *rpoB* and *MPB64* genes had higher sensitivity than *IS6110* (86% vs. 74.6%). HRM reported eight multidrug resistant (MDR), two mono-rifampicin, and five mono-isoniazid resistant cases, all were concordant with gene sequencing. qPCR followed by HRM analysis offer a simple, accurate, and rapid platform for simultaneous detection of OATB and MDR.

KEYWORDSmelt-curve analysis, multidrug resistant tuberculosis, nucleic acid amplification test, osteoarticular tuberculosis, *rpoB* gene sequencing

1 | INTRODUCTION

Rapid and accurate diagnosis of osteo-articular tuberculosis (OATB) is important to prevent bony deformities and subsequent neurological complications. Though OATB constitutes 5%–10% of all extrapulmonary tuberculosis (EPTB),¹ it is associated with high morbidity. Spinal tuberculosis, the most common form of OATB constituting nearly 50% of OATB cases, is associated with spinal deformity in nearly 50% of inadequately treated patients with 8.6% mortality.² The varied clinical and radiological features overlap with several

other diseases of the skeletal system thereby causing much confusion.³ Collecting adequate sample for microbiological diagnosis is also challenging in OATB. Further, the conventional microbiological tests are limited by paucibacillary load and low sensitivity (smear), and prolonged incubation periods (culture) required to obtain growth of *Mycobacterium tuberculosis*. As a direct consequence of these clinical and diagnostic limitations, more often than not, the patients are empirically started on antitubercular therapy (ATT) for prolonged duration beyond 6 months. This not only adds to increased drug toxicity and poor compliance, but also the emergence of drug

Kusum Sharma and Megha Sharma contributed equally as first author.

© 2022 Orthopaedic Research Society. Published by Wiley Periodicals LLC.

resistance.^{4,5} Multidrug resistant tuberculosis (MDR-TB) has been reported from various EPTB forms like meningitis,⁶ uveitis,⁷ and recently from OATB also.^{4,8,9} The situation is further compromised in high-endemic resource-limited countries like India where the facilities for drug susceptibility testing (DST) are not widely available. There is, therefore, a felt need to develop a relatively inexpensive test with wide availability that not only diagnose OATB rapidly but also detect MDR.

Phenotypic DST remains the gold standard for detecting drug resistance in TB. It requires dedicated facilities and still the results can be obtained not before 5–10 days of incubation. Molecular methods have revolutionized detection of drug resistance in TB by their speed and accuracy. Xpert and Xpert Ultra MTB/RIF assays, semi-automated platforms, were recommended by WHO in 2020 for the simultaneous detection of *M. tuberculosis* and rifampicin resistance (RifR) in EPTB samples but they do not detect isoniazid resistance (InhR).¹⁰ MTBDRplus assay offers detection of both rifampicin and InhR but is technically more demanding and labor-intensive.⁷ High-resolution melt-curve analysis (HRM) offers a middle path between these two assays by allowing detection of both rifampicin and InhR in a simpler closed-system that prevents cross-contamination and requires no postamplification processing.¹¹ In HRM, point mutations in the sample are detected based on the differences in melting temperatures between the sample DNA and reference DNA. The gene of interest is first amplified using real-time quantitative polymerase chain reaction (qPCR) followed by addition of immuno-fluorescent dyes that bind to ds-DNA. The differences in the melting temperatures of the two DNAs, as produced during the strand-displacement of denaturation step, reveal the presence of mutation.¹¹

HRM has been used earlier in detecting drug resistance in pulmonary TB (PTB). A recent systematic review evaluating >4700 PTB isolates reported that the cumulative sensitivity of HRM (using variable gene targets) in detecting RifR and InhR was 94% and 93%, respectively.¹² Limited studies are available on evaluation of HRM on EPTB samples. To the best of our knowledge, other than two studies (one on tuberculous meningitis¹³ and other on EPTB samples containing <20 OATB samples⁶), no study has yet evaluated HRM for detection of drug resistance in substantial number of OATB cases. The current study was therefore planned to evaluate potential of *rpoB* and *katG* HRM in detecting RifR and InhR, respectively, on a large cohort of OATB. The same *rpoB* gene was used for the initial amplification using qPCR and its performance was also compared with two other gene targets—*IS6110* and *MPB64*.

2 | MATERIALS AND METHODS

2.1 | Study setting

250 specimens (synovial fluid and pus aspirated from bone) processed in the Mycobacteriology division of the Department of Medical Microbiology at PGIMER, Chandigarh between 2015 and

2021 were included in the study. The study was approved by the Institute Ethics committee. These 250 specimens were divided into the following three groups:

- Group I—Confirmed OATB ($n = 20$), culture positive/smear positive/histopathology positive (epithelioid cell granuloma with caseating necrosis and acid-fast bacilli).
- Group II—Suspected OATB ($n = 130$), culture/smear negative; histopathology inconclusive (epithelioid cell granuloma but without caseation and Acid-fast bacilli) and showing clinical response to ATT, as described earlier.¹⁴
- Group III—Control group ($n = 100$), patients with inflammatory joint conditions and proven non-tubercular infectious etiology.

2.2 | Specimen processing

Standard mycobacteriological precautions were followed while processing the specimens. The pus specimens were treated with NaLC-NaOH method for decontamination and digestion. Each specimen was subjected to Ziehl Neelsen smear, Lowenstein Jensen culture, and DNA extraction. The remaining specimen was stored at -20°C for future reference. The culture-positive specimens were subjected to phenotypic DST using 1% proportion method for rifampicin (40 $\mu\text{g/ml}$) and isoniazid (0.2 $\mu\text{g/ml}$). All specimens were coded and randomly placed to blind the investigator of the grouping of patients.

2.3 | DNA extraction and qPCR

The cetyl trimethyl-ammonium bromide-phenol chloroform method was used for extracting DNA. qPCR was performed on Light Cycler 480 real-time PCR system (Roche Applied Sciences) using three different gene targets—*rpoB*, *IS6110*, and *MPB64*, as described previously.⁶ A specimen was considered positive for OATB by qPCR if positive amplification was achieved in the form of a sigmoid curve with a Ct value between 18 and 35 in any of the three targets.

2.4 | HRM analysis for RifR and InhR

All DNA positive by qPCR were subjected to HRM analysis of *rpoB* gene and *katG* gene using Light Cycler 480 software v1.5, as previously described.⁶ Briefly, a pre-incubation was given at 95°C for 1 min followed by 40°C for 1 min. Then melting was achieved by increasing temperature back to 95°C at 0.1°C/s . The software detected the decrease in fluorescence with reference to susceptible strain (*M. tuberculosis* H37Rv) and depicted it graphically as the difference in melting temperatures (T_m).

The turn-around time and expenditure (excluding workspace and personnel) of qPCR-HRM was also evaluated.

2.5 | Gene sequencing

All specimens that reported RifR and InhR using HRM and representative 30 specimen reported susceptible were subjected to *rpoB* and *katG* gene sequencing using Big-Dye 3.1 Terminator and ABI 3130 sequencer (Applied Biosystems).

2.6 | Reference standards

Two standards were used for diagnosis of OATB by qPCR; culture was used as the reference for Group I and composite reference standard¹⁴ (comprising of clinical, radiological, microbiological, and histopathological features along with response to antitubercular therapy) was used as reference for Group II. For evaluating drug resistance by HRM, phenotypic DST was used as reference for Group I and gene sequencing for both Groups I and II.

2.7 | Statistical analysis

The statistical analyses were done using Epitools software (available at Ausvets.com.au). The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were calculated for each gene target. The gene with highest sensitivity was also evaluated for its diagnostic performance using receiver operating curve (ROC) analysis and area under the curve (AUC) analysis.

3 | RESULTS

3.1 | Performance of *rpoB* qPCR in diagnosing OATB (Figure 1)

- Group I—Out of 20 confirmed OATB cases, the three genes, *rpoB*, *MPB64*, and *IS6110* detected 19, 19, and 17 cases, respectively,

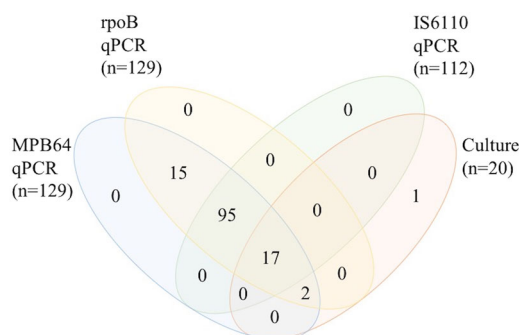


FIGURE 1 Venn diagram showing the number of OATB detected by the four tests—culture, *rpoB* qPCR, *MPB64* qPCR, and *IS6110* qPCR. OATB, osteo-articular tuberculosis; qPCR, quantitative polymerase chain reaction.

producing a sensitivity of 95%, 95%, and 85%, respectively. There was one case which was missed by both *rpoB* and *MPB64*.

- Group II—Out of 130 suspected cases, the three genes, *rpoB*, *MPB64*, and *IS6110* detected 110, 110, and 95 cases, respectively, producing a sensitivity of 84.6%, 84.6%, and 73.1%, respectively. *rpoB* missed the same 20 cases that were missed by *MPB64* also.

The ROC of *rpoB* qPCR in diagnosing OATB is presented in Figure 2 and the AUC was 0.98 (95% confidence interval: 0.98–0.996).

Overall, the sensitivity of diagnosing OATB was 86% (129/150) for both *rpoB* and *MPB64*, and 74.6% (112/150) for *IS6110*. None of the cases from Group III were positive by any of the gene targets, giving a specificity of 100% for all three genes (Table 1).

3.2 | Performance of HRM in detecting RifR and InhR (Table 2)

Group I—Among 20 confirmed cases, four (21%), two (10.5%), and five (26.3%) cases were reported as MDR, mono-RifR, and mono-InhR, respectively, by HRM. These results were 100% concordant with both phenotypic DST and gene sequencing and no false-resistance or false-susceptible was recorded.

Group II—Among 130 suspected OATB cases, four (3.6%) cases of MDR were reported by HRM. All these four cases also had mutations in both *rpoB* and *katG* genes on sequencing.

Overall, out of 129 cases detected by *rpoB* qPCR, eight were reported as MDR-OATB (both RifR and InhR), two as mono-RifR, and five as mono-InhR by HRM. The results of HRM were 100% concordant with phenotypic Drug Susceptibility Testing (pDST) and sequencing, hence the rate of MDR-OATB was 5.33% (8/150). Mutations were observed at codon 531, 533, 526, and 516 of *rpoB* gene and codon 315 of *katG* gene (Table 3).

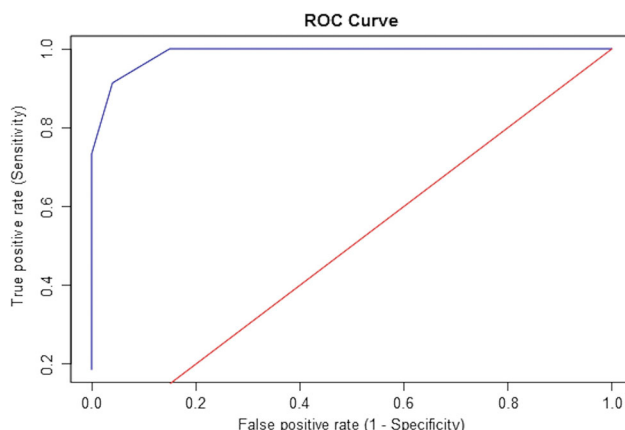


FIGURE 2 The receiver operating curve (ROC) analysis of *rpoB*-qPCR in diagnosing OATB. OATB, osteo-articular tuberculosis; qPCR, quantitative polymerase chain reaction. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Performance of qPCR using the three gene targets (*rpoB*, *MPB64*, and *IS6110*) for diagnosis of OATB

Test	Test results	OATB (N = 150)	Control group (N = 100)	Sensitivity	Specificity	PPV	NPV
Any of three qPCR	Positive	129	-100	86%	100%	100%	86.24%
	Negative	21					
<i>rpoB</i> qPCR	Positive	129	-100	86%	100%	100%	86.24%
	Negative	21					
<i>MPB64</i> qPCR	Positive	129	-100	86%	100%	100%	86.24%
	Negative	21					
<i>IS6110</i> qPCR	Positive	112	-100	74.66%	100%	100%	74.46%
	Negative	38					

Abbreviations: NPV, negative predictive value; OATB, osteo-articular tuberculosis; PPV, positive predictive value; qPCR, quantitative polymerase chain reaction.

TABLE 2 Results of HRM analysis of *rpoB* and *katG* gene for Rifampicin and isoniazid drug susceptibility testing

Group	No. of patients	No of OATB cases by qPCR	MDR N (%)	<i>rpoB</i> mono-RifR+ (%)	<i>katG</i> mono-InhR+ (%)	RifS+ (%)	InhS+ (%)
Group-1 confirmed OATB	20	19 (95)	4 (21)	2 (10.52)	5 (26.31)	13 (68.42)	10 (52.63)
Group-2 suspected OATB	130	110 (84.61)	4 (3.63)	-	-	106 (96.36)	106 (96.36)
Total	150	129 (86)	8 (6.20)	2 (1.55)	5 (3.87)	119 (92.24)	116 (89.92)
Group-3 non TB control	100	-	-	-	-	-	-

Abbreviations: InhR, isoniazid resistant; InhS, isoniazid susceptible; MDR, multidrug resistant; OATB, osteo-articular tuberculosis; qPCR, quantitative polymerase chain reaction; RifR, rifampicin resistant; RifS, rifampicin susceptible.

TABLE 3 Comparison of phenotypic susceptibility (1% proportion method), HRM, and sequencing of *rpoB* and *katG* gene for the screening of drug resistance

Drug	Target	1% proportion (N)	HRM (N)	Sequencing (N)	Mutations at codon (n; %)	Mutations
Rifampicin	<i>rpoB</i>	6	10	10	531 (6; 60%)	TCG-TTG
					533 (2; 20%)	GAC-GTC
					526 (1; 10%)	CAC-GAC
					516 (1; 10%)	CTG-CTT
Isoniazid	<i>katG</i>	9	13	13	315 (13; 100%)	AGC-ACC

Abbreviation: HRM, high-resolution melt-curve analysis.

3.3 | Cost and time analysis

The overall cost of qPCR-HRM was \$2/isolate and the results were available within 90 min.

4 | DISCUSSION

Tuberculosis is an ancient disease, and OATB is probably the oldest documented form of tuberculosis, as evidenced from the skeletal remains of ancient civilizations.¹⁵ It is unfortunate OATB continues to handicap us in the modern era. The low bacillary load,

difficult sampling, and small volume of sample are inherent limitations of OATB. Hence, there is a need for a test that could make best use of the available sample and detect both OATB and MDR (and not only RifR) with accuracy and speed. In the present study, we evaluated if qPCR followed by HRM could fill this void in the diagnosis of OATB.

The qPCR used in the current study detected OATB with a sensitivity and specificity of 86% and 100%, respectively. The gene targets of *rpoB* and *MPB64* performed equally good and outperformed *IS6110* by detecting additional 17 cases (2 confirmed and 15 suspected). This was in concordance with our previous observation wherein performance of both *rpoB* and *MPB64* in

detecting EPTB was similar (sensitivity 83%) and better than *IS6110* (sensitivity 73%).⁶ There are only a few studies evaluating qPCR for OATB. Even in those studies, the diagnosis of OATB was limited to either a single case or a single gene target,^{16,17} or studies where OATB specimens formed a small chunk of the total EPTB specimens.⁶ Our observation on performance of different gene targets is in contrast to a recent study by Antonello et al.,¹⁸ wherein in-house droplet-digital PCR (on 89 Italian EPTB specimens including four OATB specimens) reported a sensitivity of 66.2% for *rpoB* and 98.5% for *IS6110*. These differences in performance of *rpoB* could be attributed to different molecular chemistries used and copy numbers of the two genes (*rpoB* single copy number, *IS6110* upto 25). More importantly, genomic differences in *IS6110* distribution between Italian and Indian mycobacterial strains could have led to such different observations since 10%–40% of Indian mycobacterial isolates either lack *IS6110* altogether or possess only a single copy.¹⁹

We documented the rate of MDR-OATB as 5.33% ($n = 8$) by HRM in the present study; this lies within the range of 4.6%–15.1% documented in previous studies using pDST for diagnosing MDR-OATB, mostly spinal tuberculosis.^{4,8,20–24} Interestingly, other than a single study from Mumbai reporting MDR in spinal tuberculosis in 43.6% (110/252),⁸ rest all have reported MDR rate of <15%. A study by Arockiraj et al.⁴ estimated the prevalence of MDR-OATB to be 4.9% ($n = 36$) from 730 culture/Xpert confirmed cases of spinal tuberculosis in South India. Another study from Mumbai reported 13.1% MDRs from 686 cases of culture-confirmed TB spondylitis.²² All the previous reported studies, however, made use of phenotypic methods like mycobacterial growth indicator tubes for DST and were hence not only limited to specialized centers but also caused inadvertent delay in initiation of therapy for MDR. In a study by Li et al.,²¹ a mean delay of 8.4 ± 2.6 months was observed from the time of diagnosing OATB (spinal form) to obtaining results of pDST. This delay could lead to subsequent skeletal and/or neurological deficits in patients of MDR-OATB, as observed in previous studies documenting clinical follow-up of MDR-OATB cases.^{25,26} This signifies the importance of simple and rapid molecular tests that allow detection of MDR cases in a single step.

The Xpert MTB/RIF assay may have revolutionized diagnosis of PTB and RifR, but performance of Xpert was inferior to qPCR in diagnosing OATB.¹⁸ Even the WHO guidelines of 2020 that recommended Xpert Ultra as the initial test for diagnosing EPTB and detecting RifR, cautioned that the recommendation for OATB (synovial fluid) was more conditional as the evidence for test accuracy was of “low” certainty.¹⁰ Not only this, Xpert and Xpert ultra are limited in their performance as they detect only RifR and miss InhR. In the present study also, there were five cases of mono-InhR which would have been altogether missed by Xpert. Though detection of RifR alone is taken as a surrogate marker for MDR-TB and does not have negative implications on patient management, documentation of resistance to other drugs, including isoniazid and even rest of ATT should be attempted for epidemiological surveillance and control efforts. Though the HRM assay evaluated in the present study was used for detecting RifR and InhR only, it can be extrapolated for detecting resistance to other drugs used in treatment of tuberculosis.

This was the pioneer study wherein HRM was used for detecting MDR-OATB and it gave 100% concordant results with pDST and gene sequencing. There were no cases reported as false-resistance or false-susceptible. Prior studies have also reported high sensitivity and specificity for HRM, but there were false-RifR and false-RifS cases also. Arefzahde et al.,²⁷ in the analysis on 80 PTB specimens, reported that the sensitivity and specificity of HRM for determining RifR was 100% and 89.3%, respectively, with false-RifR reported in eight cases. Galarza et al.²⁸ reported a sensitivity and specificity of 98.7% and 97.7%, respectively, for HRM in determining RifR in Peruvian PTB isolates. A recent meta-analysis evaluating performance of HRM in determining drug resistance in >4700 PTB isolates reported a sensitivity and specificity of 94% and 97%, respectively, for Rif and 93% and 98%, respectively, for Inh.¹² Such results could be attributable to resistance arising from genes outside those targeted in these studies, for example, 5% of RifR is attributable to mutations at sites other than the *rpoB* gene as reported earlier also.²⁷ Similarly, a review on gene loci responsible for InhR reported out of >5804 global isolates of InhR-*M. tuberculosis*, mutations in *katG* gene were found in 4271 (73.5%), followed by *inhA-15* (13.5%) and *inhA-8* (1.8%).²⁹ The 100% sensitivity of RifR and InhR observed in the current study was because all our isolates had mutations within the most commonly encountered codons of *rpoB* gene and the *katG* gene targeted for them respectively, as confirmed by gene sequencing. Nonetheless, these target-specific limitations of HRM should be acknowledged and caution is advised in reporting resistance, especially mono-RifR (surrogate marker for MDR) and even mono-InhR (most commonly found resistance and associated with increased risk of progress to MDR). Cases of mono-resistance should be confirmed by phenotypic DST, as done in the present study.

A possible limitation of HRM is that it cannot detect resistance arising from other mechanisms like efflux pumps and gene loci other than those targeted. On the upside, HRM reported resistance to both Rif and Inh at an approximate expenditure of \$2 per isolate with a turn-around time of 90 min. This proved much cheaper and/or faster than Xpert (\$10, 1 h) and MTBDRplus (\$22, 24 h) in our previous experience also.⁶ Another limitation is the chance of high false-positivity arising from dead bacilli or patients undergoing treatment. For enabling differentiation between the live and dead bacilli, messenger RNA detection may be used, as described earlier.³⁰

To conclude, qPCR followed by HRM analysis could be used reliably for the simultaneous detection of OATB and MDR directly from the patient's specimen. The technique is simple, accurate, rapid, and inexpensive.

AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Material preparation and data collection were performed by Mandeep Singh Dhillon, Aman Sharma, and Kusum Sharma. Data analysis was performed by Kusum Sharma and Megha Sharma. The first draft of the manuscript was written by Megha Sharma and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ETHICS STATEMENT

The Institute Ethics Committee approved the study. Informed consent was obtained from all individual participants included in the study.

ORCID

Megha Sharma  <http://orcid.org/0000-0002-6740-0493>

REFERENCES

1. Pigrau-Serrallach C, Rodríguez-Pardo D. Bone and joint tuberculosis. *Eur Spine J*. 2013;22(suppl 4):556-566.
2. Merino P, Candel FJ, Gestoso I, Baos E, Picazo J. Microbiological diagnosis of spinal tuberculosis. *Int Orthop*. 2012;36(2):233-238.
3. Das R, Chandrasekar S, Purohit G, et al. "There is more than what meets the eye"—extensive spinal tuberculosis presenting as ocular tuberculosis. *Indian J Tuberc*. 2020;67(3):404-406. doi:10.1016/j.ijtb.2019.11.017
4. Arockiaraj J, Karthik R, Michael JS, et al. "Need of the hour": early diagnosis and management of multidrug resistant tuberculosis of the spine: an analysis of 30 patients from a 'high multidrug resistant tuberculosis burden' country. *Asian Spine J*. 2019;13(2):265-271.
5. Jain A, Rajsekaran S, Jaggi R, Myneedu V. Tuberculosis of the spine. *J Bone Joint Surg*. 2020;102A(7):617-628.
6. Sharma K, Sharma M, Singh S, et al. Real-time PCR followed by high-resolution melting curve analysis: a rapid and pragmatic approach for screening of multidrug-resistant extrapulmonary tuberculosis. *Tuberculosis*. 2017;106:56-61. doi:10.1016/j.tube.2017.07.002
7. Sharma K, Gupta A, Sharma M, et al. MTBDRplus for the rapid diagnosis of ocular tuberculosis and screening of drug resistance. *Eye*. 2017;32(2):451-453. doi:10.1038/eye.2017.214
8. Upadhyay M, Patel J, Kundnani V, Ruparel S, Patel A. Drug sensitivity patterns in Xpert-positive spinal tuberculosis: an observational study of 252 patients. *Eur Spine J*. 2020;29(7):1476-1482. doi:10.1007/s00586-020-06305-x
9. Jain A, Jaggi K, Bhayana H, Saha R. Drug-resistant spinal tuberculosis. *Indian J Orthop*. 2018;52(2):100-107.
10. World Health Organization (WHO). WHO Consolidated Guidelines on Tuberculosis Module 3: Diagnosis; Rapid Diagnostics for Tuberculosis Detection. World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO p.
11. Montgomery JL, Sanford LN, Wittwer CT. High-resolution DNA melting analysis in clinical research and diagnostics. *Expert Rev Mol Diagn*. 2010;10(2):219-240.
12. Keikha M, Karbalaie M. High resolution melting assay as a reliable method for diagnosing drug-resistant TB cases: a systematic review and meta-analysis. *BMC Infect Dis*. 2021;21(1):1-10. doi:10.1186/s12879-021-06708-1
13. Sharma K, Modi M, Kaur H, Sharma A, Ray P, Varma S. RpoB gene high-resolution melt curve analysis: a rapid approach for diagnosis and screening of drug resistance in tuberculous meningitis. *Diagn Microbiol Infect Dis*. 2015;83(2):144-149. doi:10.1016/j.diagmicrobio.2015.06.010
14. Vadwai V, Boehme C, Nabeta P, Shetty A, Alland D, Rodrigues C. Xpert MTB/RIF: a new pillar in diagnosis of extrapulmonary tuberculosis? *J Clin Microbiol*. 2011;49(7):2540-2545.
15. Jain AK. Tuberculosis of the spine: a fresh look at an old disease. *J Bone Joint Surg Br*. 2010;92(7):905-913.
16. Saboe A, Sakasasmitha S, Hartantri Y, et al. A case of endocarditis and spondylodiscitis associated with *Mycobacterium tuberculosis*. *IDCases*. 2021;26:e01313. doi:10.1016/j.idcr.2021.e01313
17. Colmenero JD, Morata P, Ruiz-Mesa JD, et al. Multiplex real-time polymerase chain reaction: a practical approach for rapid diagnosis of tuberculous and brucellar vertebral osteomyelitis. *Spine (Phila Pa 1976)*. 2010;35(24):1392-1396.
18. Antonello M, Scutari R, Lauricella C, et al. Rapid detection and quantification of *Mycobacterium tuberculosis* DNA in paraffinized samples by droplet digital PCR: a preliminary study. *Front Microbiol*. 2021;12:727774.
19. Chauhan DS, Sharma VD, Parashar D, et al. Molecular typing of *Mycobacterium tuberculosis* isolates from different parts of India based on IS6110 element polymorphism using RFLP analysis. *Indian J Med Res*. 2007;125(4):577-581.
20. Bhosale S, Prabhakar A, Srivastava S, Raj A, Purohit S, Marathe N. Pattern of drug resistance in primary spinal tuberculosis: a single-center study from India. *Glob Spine J*. 2021;11(7):1070-1075.
21. Li L, Zhang Z, Luo F, et al. Management of drug-resistant spinal tuberculosis with a combination of surgery and individualised chemotherapy: a retrospective analysis of thirty-five patients. *Int Orthop*. 2012;36(2):277-283.
22. Mohan K, Rawal S, Pawar UM, et al. Drug resistance patterns in 111 cases of drug-resistant tuberculosis spine. *Eur Spine J*. 2013;22(suppl 4):8-10.
23. Lan X, Xu JZ, Liu XM, Ge BF. Drug susceptibility testing guided treatment for drug-resistant spinal tuberculosis: a retrospective analysis of 19 patients. *Int Surg*. 2013;98(2):175-180.
24. Pawar UM, Kundnani V, Agashe V, Nene A, Nene A. Multidrug-resistant tuberculosis of the spine-is it the beginning of the end? A study of twenty-five culture proven multidrug-resistant tuberculosis spine patients. *Spine (Phila Pa 1976)*. 2009;34(22):806-810.
25. Arockiaraj J, Robert M, Rose W, Amritanand R, David KS, Krishnan V. Early detection and analysis of children with multidrug-resistant tuberculosis of the spine. *Asian Spine J*. 2019;13(1):77-85.
26. Seddon JA, Donald PR, Vlok GJ, Schaaf HS. Multidrug-resistant tuberculosis of the spine in children-characteristics from a high burden setting. *J Trop Pediatr*. 2012;58(5):341-347.
27. Arefzadeh S, Azimi T, Nasiri MJ, et al. High-resolution melt curve analysis for rapid detection of rifampicin resistance in *Mycobacterium tuberculosis*: a single-centre study in Iran. *New Microbes New Infect*. 2020;35:100665. doi:10.1016/j.nmni.2020.100665
28. Galarza M, Fasabi M, Levano KS, et al. High-resolution melting analysis for molecular detection of multidrug resistance tuberculosis in Peruvian isolates. *BMC Infect Dis*. 2016;16(1):1-6. doi:10.1186/s12879-016-1615-y
29. Valafar SJ. Systematic review of mutations associated with isoniazid resistance points to continuing evolution and subsequent evasion of molecular detection, and potential for emergence of multidrug resistance in clinical strains of mycobacterium tuberculosis. *Antimicrob Agents Chemother*. 2021;65(3):3-6.
30. Sharma K, Gupta A, Sharma M, et al. Detection of viable *Mycobacterium tuberculosis* in ocular fluids using mRNA-based multiplex polymerase chain reaction. *Indian J Med Microbiol*. 2022;40:254-257. doi:10.1016/j.ijmm.2021.12.019

How to cite this article: Sharma K, Sharma M, Sharma A, Dhillon MS. Diagnosing osteo-articular tuberculosis and multidrug resistance using real-time polymerase chain reaction and high-resolution melt-curve analysis. *J Orthop Res*. 2022;1-6. doi:10.1002/jor.25410